काशी हिन्दू विश्वविद्यालय

BANARAS HINDU UNIVERSITY

Mobile Application Development

Bachelor of Science – 2nd Semester

By

Shashvat Singh

Department of Computer Science
Banaras Hindu University

```
fun main() {
    addition(5,10)
    addition(15,20)
    addition(25,50)
}
fun addition(a:Int, b:Int){
    var c: Int = a + b
    println("$c")
}
```

```
15
35
75
```

Working with Image ( )

This is a composable function used to display images in Jetpack Compose.

Working with Image ( )

Save your Image to given Directory:

C: > Users > shashvat > StudioProjects > FirstApplication > app > src > main > res > drawable > myimage.jpg

It will be treated as R.drawable.myimage

Working with Image ( )

R.drawable.shashvat (if image name is shashvat.jpg)

```
Image(
    painter = painterResource(id = imageResId),
    contentDescription = null,
    contentScale = ContentScale.Crop,
    modifier = Modifier
        .size(56.dp)
        .clip(CircleShape)
        .background(Color.Gray)
)
```

Parameters:

1. painter = painterResource (id = imageResId)

- painterResource( ): Loads an image from a **drawable** resource using its ID (imageResId).
- painter is the actual image resource that will be painted on the screen.

2. contentDescription = null

- This provides a description of the image for accessibility services (like screen readers).
- null means the image is decorative and shouldn't be described for accessibility purposes.

3. contentScale = ContentScale.Crop

- Defines how the image should be resized to fit the space.
- ContentScale.Crop: Scales the image so that it fills the target bounds and crops the extra parts that don’t fit.

4. modifier = Modifier

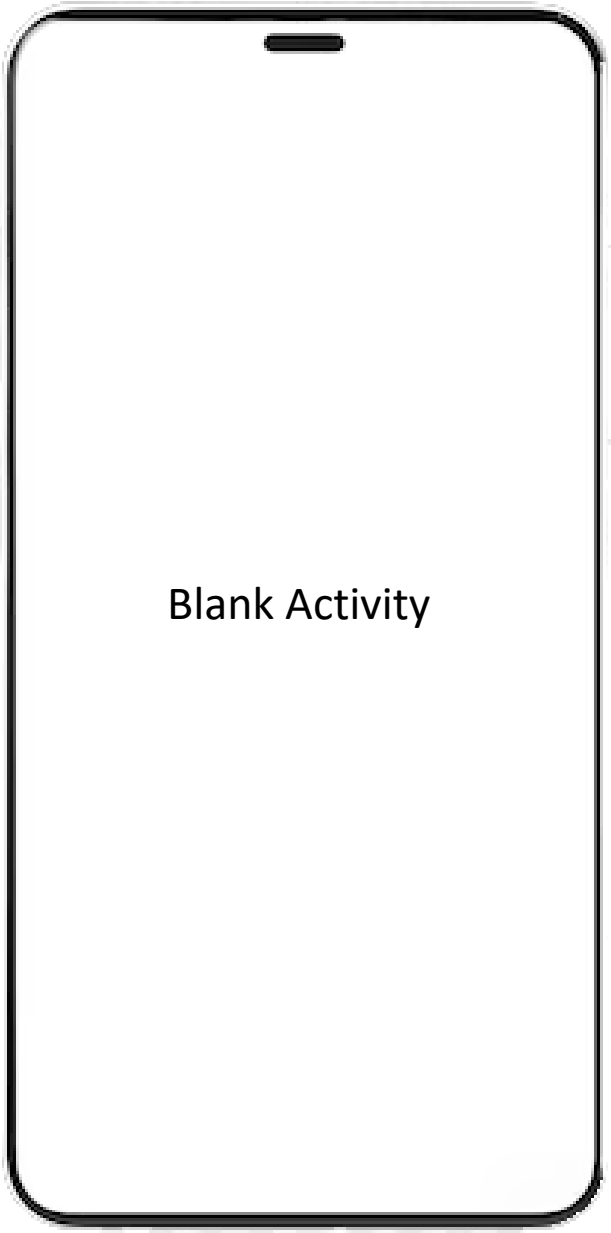
Objectives

Activity 1:

- An Activity which contains list of Users (row-wise).
- Each Row contains Name and Profile picture of the user.

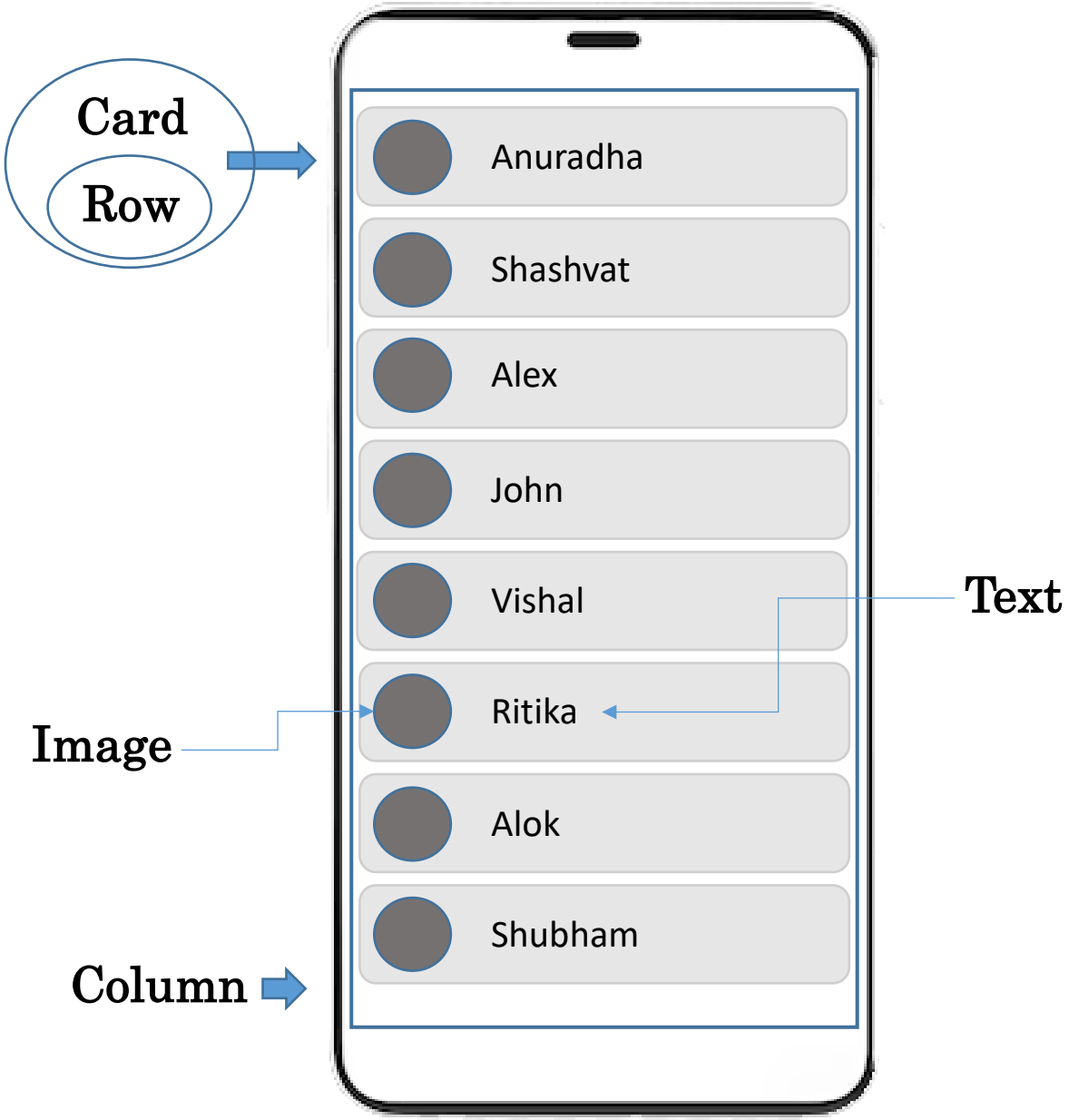
Activity 2:

- This activity contains the detailed information of respective user.
- A Back button to go to *Activity 1.*



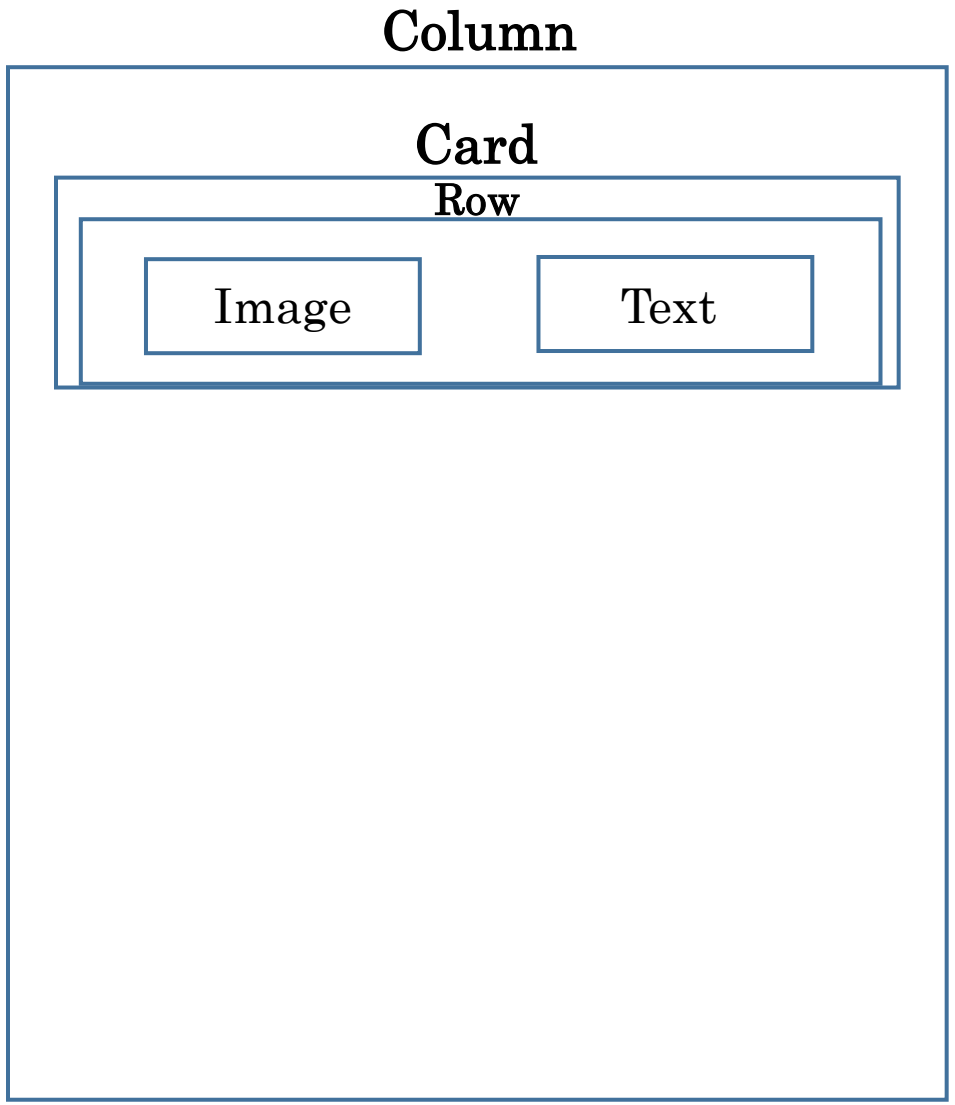
Activity 1

- An Activity which contains list of users (row-wise).
- Each Row contains Name and Profile picture of the user



Activity 1

- An Activity which contains list of users (row-wise).
- Each Row contains Name and Profile picture of the user



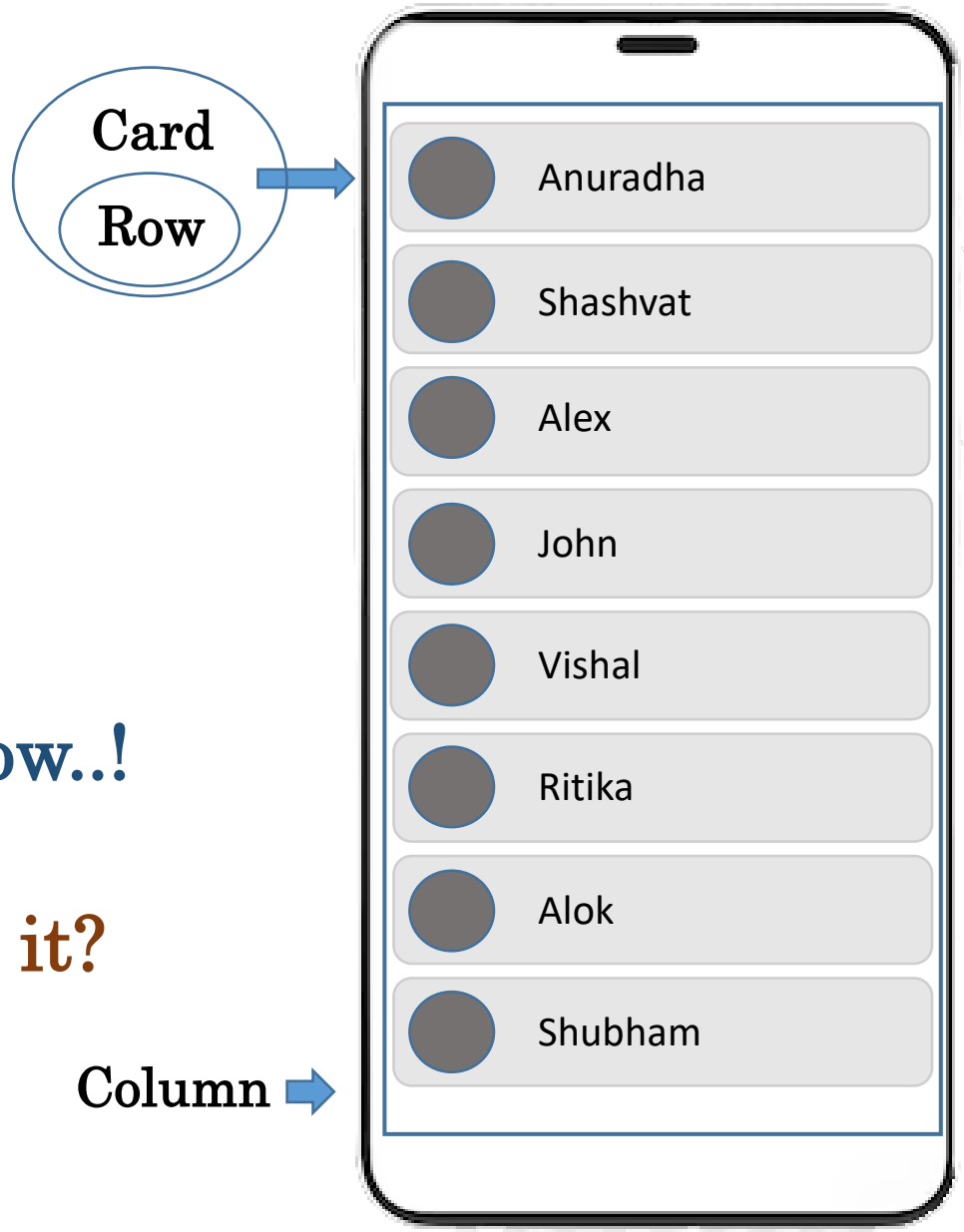
Activity 1:

Each row Requires three data:

1. Name
2. Image
3. Related Activity (Detailed view)

These data will be repeated in each row..!

Can we create a common function for it?



```
fun main() {
    addition(5,10)
    addition(15,20)
    addition(25,50)
}
fun addition(a:Int, b:Int){
    var c: Int = a + b
    println("$c")
}
```

```
15
35
75
```

Activity 1: The CODES

Two functions will be required

1. **UserCard( )**
   To display a row with username and image.

2. **UserListScreen( )**
   To call UserCard function multiple times to display multiple rows of users.

Activity 1: The CODES

UserCard function

```
@Composable
fun UserCard(name: String, imageResId: Int, onClick: () -> Unit) {}
```

The CODES

UserCard function + UI element *Card*

```
@Composable
fun UserCard(
    name: String,
    imageResId: Int,
    onClick: () -> Unit) {
    Card(
        modifier = Modifier
            .fillMaxWidth()
            .height(80.dp)
            .clickable { onClick() }, // clickable here
        elevation = CardDefaults.cardElevation(6.dp),
        colors = CardDefaults.cardColors(Color.White)
    ) {
```

The CODES

UserCard function + UI element *Card + Rows*

```
Card(
    modifier = Modifier
        .fillMaxWidth()
        .height(80.dp)
        .clickable { onClick() }, // clickable here
    elevation = CardDefaults.cardElevation(6.dp),
    colors = CardDefaults.cardColors(Color.White)
) {
    Row(
        verticalAlignment = Alignment.CenterVertically,
        modifier = Modifier
            .fillMaxSize()
            .padding(12.dp)
    ) {
```

Activity 1: **The CODES**

UserCard function
+
UI element *Card*
+
Rows
+
Image and Text

```
Row(
    verticalAlignment = Alignment.CenterVertically,
    modifier = Modifier
        .fillMaxSize()
        .padding(12.dp)
) {
    Image(
        painter = painterResource(id = imageResId),
        contentDescription = null,
        contentScale = ContentScale.Crop,
        modifier = Modifier
            .size(56.dp)
            .clip(CircleShape)
            .background(Color.Gray)
    )
    Spacer(modifier = Modifier.width(16.dp))
    Text(
        text = name,
        fontSize = 18.sp,
        color = Color.Black
    )
}
```

Complete Code UserCard

```
@Composable
fun UserCard(
    name: String,
    imageResId: Int,
    onClick: () -> Unit) {
    Card(
        modifier = Modifier
            .fillMaxWidth()
            .height(80.dp)
            .clickable { onClick() }, // clickable here
        elevation = CardDefaults.cardElevation(6.dp),
        colors = CardDefaults.cardColors(Color.White)
    ) {
        Row(
            verticalAlignment = Alignment.CenterVertically,
            modifier = Modifier
                .fillMaxSize()
                .padding(12.dp)
        ) {
            Image(
                painter = painterResource(id = imageResId),
                contentDescription = null,
                contentScale = ContentScale.Crop,
                modifier = Modifier
                    .size(56.dp)
                    .clip(CircleShape)
                    .background(Color.Gray)
            )
            Spacer(modifier = Modifier.width(16.dp))
            Text(
                text = name,
                fontSize = 18.sp,
                color = Color.Black
            )
        }
    }
}
```

Activity 1: The CODES

UserListScreen function

```
@Composable
fun UserListScreen() {
    val context = LocalContext.current
    Column(
        modifier = Modifier
            .fillMaxSize()
            .padding(24.dp),
        verticalArrangement = Arrangement.spacedBy(12.dp)
    ) {
```

Activity 1: The CODES

UserListScreen function + UserCard

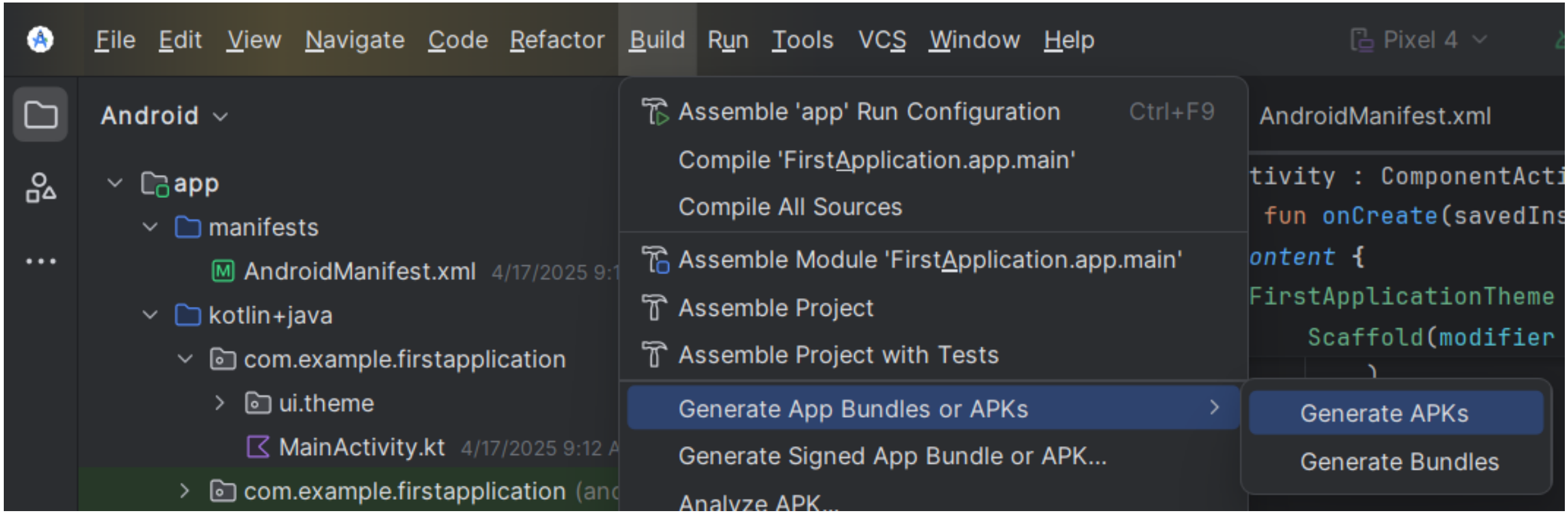
```
Column(
    modifier = Modifier
        .fillMaxSize()
        .padding(24.dp),
    verticalArrangement = Arrangement.spacedBy(12.dp)
) {
    UserCard(
        name = "Shashvat Pratap Singh",
        imageResId = R.drawable.images,
        onClick = {
            val intent = Intent(context, DetailActivity::class.java)
            intent.putExtra( name: "name",  value: "Shashvat Pratap Singh")
            context.startActivity(intent)
        }
    )
```

Multiple UserCards

```
UserCard(
    name = "Tom",
    imageResId = R.drawable.tom,
    onClick = {
        val intent = Intent(context, DetailActivity::class.java)
        intent.putExtra( name: "name",  value: "Tom")
        context.startActivity(intent)
    }
)
UserCard(
    name = "Daniel Craig",
    imageResId = R.drawable.images,
    onClick = {
        val intent = Intent(context, DetailActivity::class.java)
        intent.putExtra( name: "name",  value: "Daniel Craig")
        context.startActivity(intent)
    }
)
```

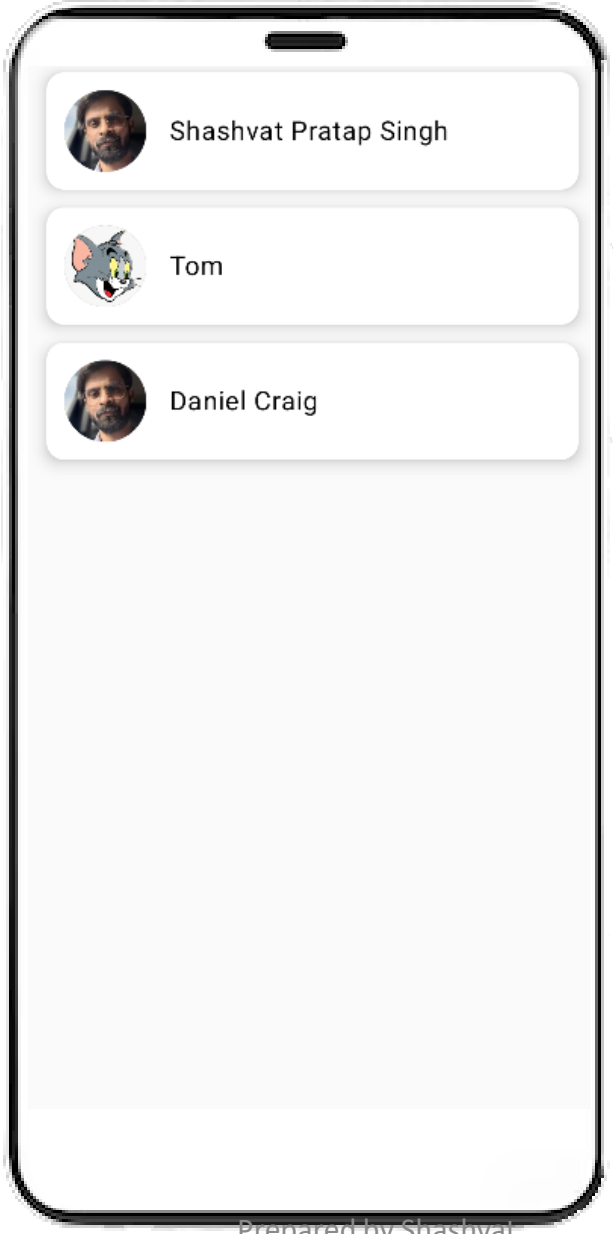
Build your First APK: Android Package Kit

Goto – Main Menu > Build > Generate App Bundles or APKs > Generate Apk



Saved -
"C:\Users\shash\StudioProjects\FirstApplication\app\build\outputs\apk\debug\app-debug.apk"

Output



THANK YOU